Bias Collapse in Shallow ReLU Networks

Numerical Evidence for Open Problems 4.1, 4.2, and 4.3

NSF RTG

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Motivation

The overparameterization puzzle:

- Modern networks have far more parameters than the problem demands
- Classical learning theory predicts overfitting; practice shows robust generalization
- Why does the optimizer self-simplify?

Bias collapse: a concrete mechanism

- Gradient flow automatically eliminates redundant neurons, with no regularization required
- $m = 1500$ neurons $\longrightarrow$ effective complexity of $k = 7$ neurons
- The number of surviving *bias clusters* equals the number of *inflection points* of the target function

This work

Rigorous numerical investigation of three open problems:

1. **OP 4.1:** Does $C(m, f^*) \to k$ as $m \to \infty$?
2. **OP 4.2:** Does collapse generalize to $d \geq 2$ dimensions?
3. **OP 4.3:** Is there a provable L^2 pruning bound after collapse?

72 runs, 9 targets, m up to 5000.

Model and Gradient Flow

1D shallow ReLU network:

$$f(x) = \sum_{j=1}^{m} a_j\, \sigma(x - b_j), \quad \sigma = \max(0, \cdot), \quad x \in [-1, 1]$$

Each neuron places a *kink* at bias b_j with slope change a_j.

Gradient flow on MSE $\mathcal{L} = \frac{1}{2}\int_{-1}^{1}(f - f^*)^2\, dx$**:**

$$\dot{a}_j = -\int_{-1}^{1}(f - f^*)\, \sigma(x - b_j)\, dx$$

$$\dot{b}_j = a_j \int_{b_j}^{1}(f - f^*)\, dx$$

Solved via `scipy` RK45, $T = 500$, rtol $= 10^{-4}$.
Stationarity criterion: $\max_j |\dot{a}_j|, |\dot{b}_j| < 0.01$.

Key measured quantity:

$$C(m, f^*) \;=\; \#\{\text{bias clusters}\}$$

at stationarity; gap threshold 0.02.

Fixed-point condition

Each *active* neuron ($a_j \neq 0$) satisfies $R_j := \int_{b_j}^{1}(f - f^*)\, dx \approx 0$ at stationarity. Verified numerically at every converged run; connects to the theory of OP 4.1.

The Three Open Problems

OP 4.1:Cluster Count

$$\lim_{m\to\infty} C(m, f^*) = k$$

Does the number of bias clusters converge to the number of sign-changing inflection points of f^*?

OP 4.2:Higher Dimensions

For $f^* : [-1, 1]^2 \to \mathbb{R}$, does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

OP 4.3:Pruning Bound

After collapse, is there a provable L^2 error bound when pruning to k neurons?

$$\left\| \tilde{f} - f \right\|_{L^2} \leq \delta \cdot \sum_j |a_j|$$

Each open problem is introduced in detail immediately before its results.

Experimental Design

Nine target functions:

$f^*(x)$	k	type
$\sin(\pi x)$	1	trig
x^3	1	poly
$\sin(2\pi x)$	3	trig
$x^5 - 3x^3$	3	poly
$\sin(3\pi x)$	5	trig
$\sin(4\pi x)$	7	trig
$\sin(5\pi x)$	9	trig
$\sin(6\pi x)$	11	trig
$\sin(7\pi x)$	13	trig

Trig: $k = 2n - 1$. Poly: irregular inflection spacing.

Main sweep (`simulate_parallel.py`):

- $m \in \{50, 100, 250, 500, 1000, 2000, 3500, 5000\}$, $T = 500$
- **72 completed runs**, 9 targets; 70/72 fully stationary $(\max |\dot{a}_j| < 0.01)$

Supporting experiments:

Experiment	Runs	Purpose
`simulate_discrete.py`	52	Discrete GD vs. ODE
`lottery_ticket_experiment.py`	52	OP 4.1.4 (survival)
`verify_pruning.py`	72	OP 4.3 bound check
`collapse_v2.py`	9	OP 4.2, 2D targets
`instability_test.py`	22	Stability near k

Open Problem 4.1: Cluster Count Conjecture

Conjecture

$$\lim_{m\to\infty} C(m, f^*) \;=\; k,$$

where $k :=$ # sign-changing inflection points of f^* in $(-1, 1)$

$C(m, f^*)$ = number of distinct bias clusters at stationarity. k = number of sign-changing inflection points of f^*.

Sub-problems we address:

- **4.1.1:** Can we prove $C(m, f^*) \le k$ independently of m?
- **4.1.4:** Which neurons survive collapse to become cluster representatives?

Why k is the right target

For $\sin(n\pi x)$, the second derivative $(f^*)'' = -n^2\pi^2 \sin(n\pi x)$ has exactly $k = 2n - 1$ sign changes in $(-1, 1)$.

At stationarity, each active neuron satisfies

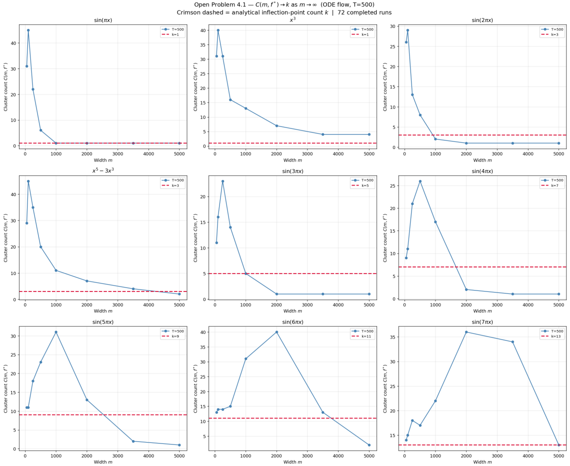
$$R_j := \int_{b_j}^{1} (f - f^*)\, dx = 0.$$

A k-cluster arrangement satisfies this condition at the inflection locations.

What makes this hard

Above-k and below-k stationary states both exist numerically. The conjecture requires showing gradient flow selects $C = k$ *from generic initializations*.

OP 4.1: $C(m, f^*)$ Across All 9 Targets



$C(m, f^*)$ vs. m at $T = 500$ (72 runs, 9 targets). **Crimson dashed line** $= k$. All panels show rise-then-fall. Low-k trig targets reach $C = k$; $\sin(7\pi x)$ ($k = 13$) reaches $C = 13$ at $m = 5000$ for the first time (stationarity unconfirmed).

OP 4.1: The Rise-Then-Fall Dynamic

Universal pattern across all 9 targets:

1. **Rise:** Small m cannot populate all k inflection attractors simultaneously. C grows above k.
2. **Peak:** C reaches a maximum scaling with k (up to $C = 40$ for $k = 11$).
3. **Fall:** With sufficient redundancy, collapse proceeds and C decreases toward k.

Key observation

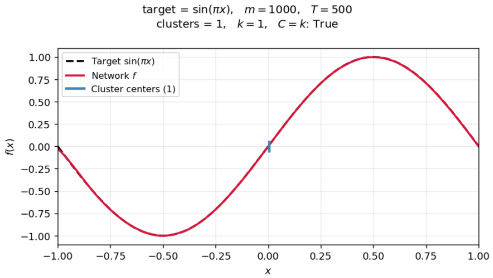
The peak of $C(m)$ shifts to **larger** m as k grows. High-k targets require much wider networks before the fall phase begins.

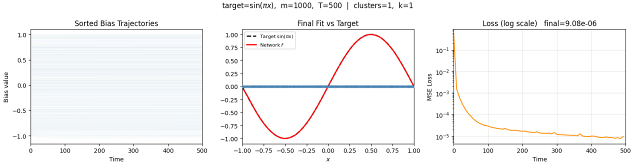
Selected $C(m)$ at $T = 500$:

Target	k	m=1000	m=2000	m=5000
$\sin(\pi x)$	1	**1**	1	1
$\sin(3\pi x)$	5	**5**	†1	†1
$\sin(4\pi x)$	7	17	2	†1
$\sin(7\pi x)$	13	22	36	‡**13**
$x^5 - 3x^3$	3	11	7	2
x^3	1	13	7	4

† Cluster structure absent or too dense for fixed threshold to detect.
‡ $C = k$ but not fully stationary ($\max|\dot{a}_j| = 0.034$).

OP 4.1 Confirmation: $\sin(\pi x)$, $k = 1$





$\sin(\pi x)$, $m = 1000$, $k = 1$

Top: Clean final fit vs. target. Blue ticks = single cluster center.

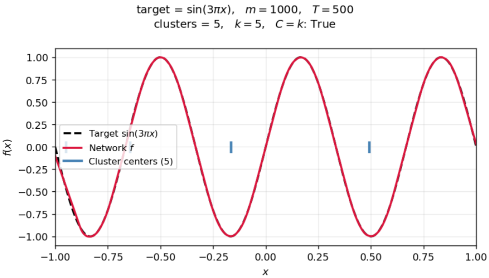
Bottom: Bias trajectories (all 1000 collapse to one location), final fit, and MSE loss on log scale.

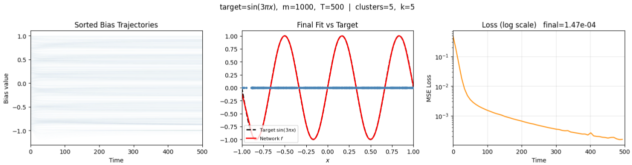
Strongest confirmation

$C = 1 = k$ for *all* $m \geq 1000$ through $m = 5000$, fully stationary ($\max|\dot{a}_j| < 3.6 \times 10^{-3}$).

Only target with $C = k$ confirmed robustly across the full m sweep.

OP 4.1 Confirmation: $\sin(3\pi x)$, $k = 5$





$\sin(3\pi x)$, $m = 1000$, $k = 5$

Top: Clean final fit. Five blue ticks mark the 5 bias clusters, one per inflection point.

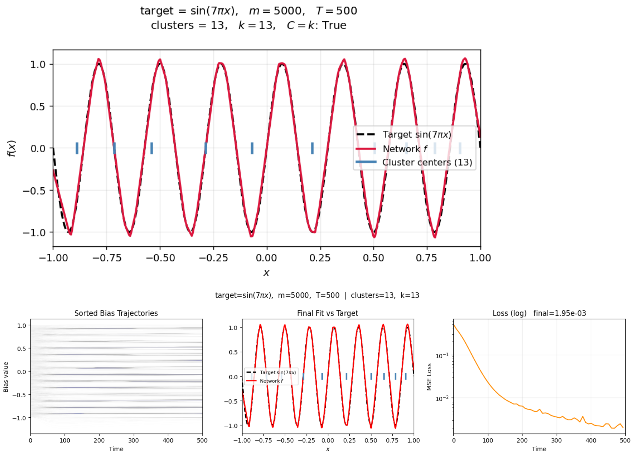
Bottom: Bias trajectories, final fit, MSE loss on log scale.

Non-monotone path to k

$$\underbrace{8}_{m=50} \rightarrow \underbrace{14}_{\text{peak},\ m=500} \rightarrow \underbrace{\mathbf{5}}_{k,\ m=1000} \rightarrow \underbrace{\dagger 1}_{m=2000}$$

Rises above k, reaches $k = 5$, then †cluster structure dissolves at larger m.

OP 4.1 New at $m = 5000$: $\sin(7\pi x)$, $k = 13$



$\sin(7\pi x)$, $m = 5000$, $k = 13$

Top: Clean final fit. Thirteen blue ticks align with all 13 inflection points.

Most complex target tested. First time $C = k = 13$ is reached.

Result

$C = 13 = k$ at $m = 5000$.

Path: $14 \to 22 \to 36 \to 34 \to \mathbf{13}$

$(m = 50, 1000, 2000, 3500, 5000)$

Caveat

$\max |\dot{a}_j| = 0.034 > 0.01$. May be **transient**; longer T needed to confirm.

OP 4.1: The Below-k Phenomenon

Three cases at verified stationary states with $C < k$. Re-run with 2 additional seeds each:

Target	k	m	seed 42	seed 5	seed 25
$\sin(2\pi x)$	3	1000	2	1	1
$\sin(4\pi x)$	7	2000	2	3	5
x^5-3x^3	3	5000	2	2	**3**$= k$

C **varies across seeds** in all three cases. No single below-k value is consistent.

Interpretation

C varies across seeds $\Rightarrow$ below-k states are **initialization-dependent local minima**, not universal fixed points.

Gradient flow has multiple basins of attraction. The k-cluster basin exists but is not the only one.

x^5-3x^3 seed 25: $C = k = 3$

First time x^5-3x^3 reaches $C = k$. Shows the k-cluster attractor exists for polynomial targets, though harder to find.

OP 4.1: Polynomial vs. Trigonometric Targets

Same k, very different convergence speed:

Target	k	m=1000	m=2000	m=5000
$\sin(\pi x)$	1	**1**	1	1
x^3	1	13	7	4
$\sin(2\pi x)$	3	2	†1	†1
x^5-3x^3	3	11	7	2

x^3 reaches only $C = 4$ at $m = 5000$, despite $k = 1$.

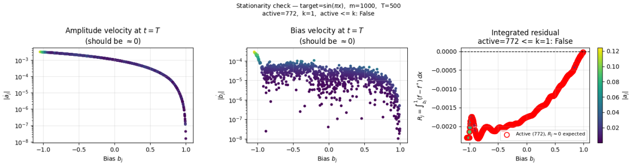
Why inflection geometry matters

For $\sin(n\pi x)$, the k inflection points are *evenly spaced* across $[-1, 1]$, forming strong, symmetric attractors that neurons organize around early in training.

Polynomial targets have *irregular* spacing: weaker, asymmetric attractors. Neurons take far longer to find them.

Implication: a proof of OP 4.1 must account for attractor geometry, not just count.

OP 4.1: Stationarity and Fixed-Point Verification



$\sin(\pi x)$, $m = 1000$, $T = 500$

Final (a_j, b_j) substituted back into the ODEs.

Left: $|\dot{a}_j| \approx 10^{-3}$–$10^{-5}$

Center: $|\dot{b}_j| \approx 10^{-4}$–$10^{-5}$

Right: $R_j \approx 0$ for active neurons (red), confirming the theoretical fixed-point condition numerically.

Discrete GD comparison

Discrete GD does *not* produce $C \to k$ for any target except $\sin(\pi x)$. The conjecture is specific to continuous ODE gradient flow dynamics.

Instability Near k: Results (36/36 complete)

Question: Are above-k states unstable? Does gradient flow dissolve extra clusters toward k?

22 qualifying runs: 8 exact_k, 8 above_k, 6 below_k. $T_{\text{perturb}} = 1000$.

Category	Status	Finding
above_k	8/8 done	**No change.** C stays fixed; ODE fixed points do not dissolve.
below_k	12/12 done	**No change.** Injected neuron amplitude stays at 0.01 and cannot grow.
exact_k	16/16 done	**All returned to k.** Exact-k states are stable attractors.

above_k: stable fixed points

Continuing ODE for $T = 1000$: C does **not** decrease. Above-k states are *stable* fixed points, not metastable.

below_k: injection-resistant

Injected neurons gain **zero amplitude**. ODE provides no restoring force toward k. Supports: below-k states are true fixed points.

Implication for OP 4.1.1

$C \to k$ cannot rely on above-k instability. Proof must use *initialization geometry*, not gradient flow instability.

Open Problem 4.2: Higher-Dimensional Collapse

Question

For a structured target $f^* : [-1,1]^2 \to \mathbb{R}$, does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

2D network:

$$f(x) = \sum_{j=1}^{m} a_j \, \sigma(w_j^\top x + b_j), \quad x \in [-1,1]^2$$

Each neuron defines an oriented hyperplane (line in 2D). Scale-invariant: $\theta_j = \angle(w_j)$, $\beta_j = b_j/\|w_j\|$.

Collapse in 2D: the (θ_j, β_j) cloud concentrates onto a low-dimensional subset.

Diagnostic

Angular entropy $H(\{\theta_j\})$: near 0 if collapsed; near $\log(36) \approx 3.58$ if uniform. Effective neurons $N_{\text{eff}} = e^H$:
$N_{\text{eff}}{=}1$ full collapse; $N_{\text{eff}}{\approx}36$ uniform.

Width sweep

$m \in \{64, 256, 512, 1024\}$, 6000 epochs, Adam.

Three targets studied

Ridge, Separable, and Radial.
See next slide for expected vs. observed behavior.

Open Problem 4.2: Target Functions

Ridge

$\sin(2\pi u^\top x) + 0.5\sin(4\pi u^\top x)$

Expected (small width):
concentration near $\theta \approx 53^\circ$

Observed:
concentration weakens as width increases

Separable

$\sin(2\pi x_1) + 0.3\sin(2\pi x_2)$

Expected (small width):
two dominant direction families

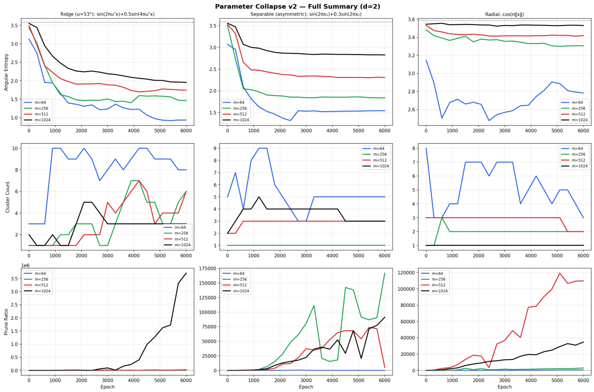
Observed:
directional structure becomes diffuse at large width

Radial (reference)

$\cos(\pi\|x\|)$

Reference case:
approximately isotropic orientation distribution

OP 4.2: Orientation Distributions



Entropy H / Eff. neurons $N_{\text{eff}}\!=\!e^{H}$ at epoch 6000:

Target	$m\!=\!64$	$m\!=\!256$	$m\!=\!512$	$m\!=\!1024$
Ridge	0.94 / 3	1.46 / 4	1.74 / 6	1.95 / 7
Separable	1.55 / 5	1.84 / 6	2.31 / 10	2.83 / 17
Radial	2.78 / 16	3.31 / 27	3.42 / 31	**3.53 / 34**

$N_{\text{eff}}\!=\!1$: full collapse; $N_{\text{eff}}\!\approx\!36$: uniform. Max $H\!=\!\log(36)\!\approx\!3.58$

Open question

Why does directional concentration weaken as width grows? Observed collapse appears strongest in moderately overparameterized networks. The mechanism behind this trend remains unclear.

OP 4.2: Key Findings

What the data shows

- At small width, directional concentration reflects target geometry:
  ridge $\to$ one direction family;
  separable $\to$ two families;
  radial $\to$ broadly distributed.
- Concentration decreases steadily as width grows.
- Networks increasingly fit targets using many distinct directions.

Effective directional complexity

$N_{\text{eff}} = e^H$ measures how many distinct directions the network actively uses.

- N_{eff} increases monotonically with width.
- Wider networks rely less on directional collapse.
- Radial $m{=}1024$: $N_{\text{eff}} \approx 34$ of 36 — nearly uniform.

Open Problem 4.3: Pruning Bound

Bound (from the slides)

$$\left\|\tilde{f} - f\right\|_{L^2} \leq \delta \cdot \sum_{j=1}^{m} |a_j|$$

Setup: After gradient flow, bias clusters have formed. *Prune* by replacing each cluster with a single representative neuron, giving $\tilde{f}$ with only k neurons.

$\delta =$ maximum intra-cluster diameter (how spread out the biases are within each cluster).

$\sum |a_j| =$ total weight mass of the full m-neuron network.

Why it matters: Provides a formal L^2 certificate that collapsing from m neurons to k neurons is safe, bridging the collapse phenomenon and approximation theory.

Why the bound holds

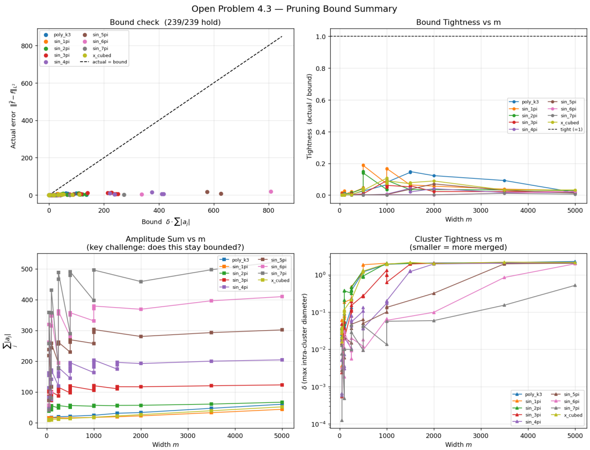
Within a cluster, all b_j differ by at most δ, so $\sigma(x-b_j) \approx \sigma(x-b_{\text{center}})$ up to an error $\leq \delta$. Summing over m neurons, errors accumulate at most as $\delta \cdot \sum |a_j|$.

The challenge

For the bound to be *non-vacuous* (goes to 0 as networks grow), we need $\delta \cdot \sum |a_j| \to 0$.

Our data shows $\sum |a_j| \sim \sqrt{m}$. So we need $\delta \to 0$ fast enough, which must come from longer training T, not wider networks m.

OP 4.3: Pruning Bound Verification



Bound: $\left\|\tilde{f} - f\right\|_{L^2} \leq \delta \cdot \sum_j |a_j|$

72 / 72 runs: bound holds

Tightness (actual / bound): 0.0002 to 0.148. Substantial slack in all cases.

Key obstacle:

- $\sum_j |a_j| \sim \sqrt{m}$ empirically
- δ does not shrink with m at $T = 500$
- Bound loosens $\sim \sqrt{m}$ while actual error plateaus

Path to non-vacuousness

$\delta \to 0$ requires $T \to \infty$, *not* $m \to \infty$. Study $\delta(T)$ at fixed m to make the bound formally useful.

OP 4.1.4 Lottery Ticket: Which Neurons Survive?

Setup: At $t = 0$, amplitudes $a_j \sim \mathcal{N}(0, 0.01)$ are random. Only the bias placement b_j is structured. Neurons near an inflection point are *geometrically lucky*.

Five conditions per (target, m): `full`, `geometric`, `bias_only`, `amp_only`, `random_k`

52 runs; 4 targets, $m \in \{500, 1000, 1500\}$.

Claim 2 (performance): FALSE

Lucky k neurons alone achieve $100\times$–$50{,}000\times$ higher loss than the full network. k kinks cannot approximate a smooth target; full redundancy is necessary.

Claim 1 (survival): bias position is key

geometric $\approx$ **bias_only** in every case (loss diff < 0.002). Initial amplitudes carry *no information*.

amp_only is the worst condition: good amplitudes with random biases performs worse than good biases with random amplitudes.

Conclusion:
Bias position at $t = 0$ is the sole informative quantity for collapse. The "winning ticket" is a good bias placement, not a good amplitude.

Summary of Evidence

Problem	Main finding	Status
OP 4.1: $C \to k$	Confirmed for $\sin(\pi x)$ ($k = 1$), all $m \geq 1000$. $\sin(7\pi x)$: $C = 13 = k$ at $m = 5000$ (stationarity unconfirmed). Below-k: **init-dependent local minima** (multiple seeds).	**Supported** for $k = 1$; **open** for $k \geq 3$
Instability near k	above_k: stable fixed points (8/8, no dissolution). below_k: injection-resistant (12/12, zero amplitude growth). exact_k: all 16 returned to k (stable attractors).	**Complete**
OP 4.2: 2D collapse	Directional concentration tracks target structure at small width. Decreases consistently with width ($m \in \{64, 256, 512, 1024\}$). Effective directional complexity (N_{eff}) increases with m.	**Partial**
OP 4.3: Pruning bound	72 / 72 hold. $\sum \lvert a_j \rvert \sim \sqrt{m}$. Non-vacuousness requires $\delta \to 0$ as $T \to \infty$.	**Verified numerically**; **open theoretically**
OP 4.1.4: Lottery ticket	Bias position at $t = 0$ is sole informative quantity. Amplitude values irrelevant. Performance claim false.	**Resolved**

Open Questions and Next Steps

OP 4.1 (open):

- A proof of $C \to k$ must account for initialization geometry. Below-k local minima exist, so the k-cluster basin must be shown to dominate at large m
- **4.1.1:** above_k states are *stable* fixed points, so instability alone cannot drive $C \to k$
- $\sin(7\pi x)$**:** Re-run at $T = 1000$ to confirm stationarity of $C = 13$

OP 4.2 (open):

- Why does directional concentration weaken as width grows? The mechanism behind this trend remains unclear.

OP 4.3, path to a formal proof:

- Study $\delta(T)$ at fixed m: does $\delta \to 0$ as $T \to \infty$?
- Exponent α in $\sum_j |a_j| = O(m^\alpha)$; data suggests $\alpha \approx 1/2$
- If $\delta = O(T^{-\gamma})$ the bound becomes non-vacuous as $T \to \infty$

Immediate priorities

(1) $\sin(7\pi x)$ at $T = 1000$ (confirm $C = 13$)
(2) $\delta(T)$ trajectory study
(3) Theory for capacity-constrained collapse

Conclusion

What we showed:

- $C \to k$ confirmed for $\sin(\pi x)$ ($k = 1$), $m \geq 1000$. First high-k evidence: $\sin(7\pi x)$ reaches $C = 13$ at $m = 5000$.
- **Rise-then-fall** is universal: peak scales with k; inflection geometry sets the convergence rate.
- **Below-k:** 3 targets have $C < k$ at genuinely stationary states, an open challenge for OP 4.1.
- **Instability:** Above-k states are stable fixed points. Proof of $C \to k$ must use initialization geometry.
- **OP 4.2:** Effective directional complexity grows with width; wider networks rely less on orientation collapse than narrower ones.
- **OP 4.3:** Bound holds 72/72; $\delta \to 0$ with T is the path to a formal proof.
- **Lottery ticket:** bias position at $t = 0$ is the sole informative quantity.

Broader implication

Gradient flow is not only an optimizer; it is a **compression algorithm** that discovers the minimum-complexity representation of the target's intrinsic structure.

Inflection points of f^* act as **fixed-point attractors**: neurons near one survive; the rest dissolve.

A rigorous proof would explain generalization in overparameterized networks.

Thank you

Supported by NSF RTG

Repository & documentation `github.com/calebabr/NSF-RTG-Project4` — full scripts, data, and result tables in `README.md`